

ZS-F32

The Conditional-Expectation Lift of the Z-Spin Continuous Core: Operator Structure, the Loxodromic Seam, and the Compact-Monodromy Reduction of the Cosmological Frontier to a Single Odd-Sector Susceptibility

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Verification: 18 PROVEN/VERIFIED checks · 7 IMPORTED-PROVEN theorem-conditions · 4 numerical regressions · 1 unresolved absolute scale (χ^-) + 2 registered conditional bridge gates | Zero new fitted parameters | $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED

§0. Abstract

This is the consolidated final version of ZS-F32. It is self-contained: every result and registered gate F32.1–F32.27 is restated with its hypotheses, conclusion, a short proof sketch, and a falsification condition, indexed in a single Results-Architecture table (§1.3). It does *not* derive the observed cosmological constant; it proves and constructs the operator-algebraic, Markovian, modular, and loxodromic structure that reduces the absolute-scale problem to a UV odd-sector susceptibility, and it isolates the residual honestly.

Operator core. A real-time modular automorphism cannot carry the i-tetration contraction (F32.1); the population-fixing phase-covariant generators realizing the time-one map λ^* form a complete integer-indexed (countable logarithmic-branch) family (F32.2), with the principal branch $n=0$ minimal in norm since $0 < \omega < \pi$ (F32.3); the seam inverts the frequency (F32.4) and the Koopman tower closes in symmetric Fock space with $\dim Z=2$ (F32.5). The 121 equal-trace face projections generate a *finite abelian* face algebra $A_{\text{face}} \cong \mathbb{C}^{121}$ that is **not** maximal abelian in the Type II_1 observer factor (F32.6), with a canonical conditional expectation E_{face} (F32.7) kept *distinct* from the non-commutative Wirth register lift E_{reg} (F32.8–F32.9). The bulk–boundary entropy production intertwines (F32.10); the seam element is loxodromic, not a pure boost (F32.11–F32.12); and the observer-corner equality is a Type I/II_1 no-go replaced by a finite coarse-graining (F32.17–F32.18).

Cosmological reduction. Even vacuum loops (incl. gravitons) are sequestered (F32.13); a single loxodromic pulse fails (F32.19) but two same-axis pulses reconstruct U_Z exactly (F32.20–F32.21, numerically robust under the Reuleaux boundary); parity alone does not isolate the odd flux (F32.22) and the three flux routes do not select its value (F32.23). Promoting F_- to a compact $U(1)$ 3-form gives Dirac-quantized flux and monodromy protection (F32.24). **Under a canonical quadratic / large-N branch law and the conditional holonomy identification $\theta Z = \omega$,** the unique branch is $k^* = 0$ (F32.25), confirmed spectrally on the compact flux circle (F32.26), giving $\mathbf{p}_{\Lambda, Z} = \frac{1}{2} \chi^- \omega^2$ with $\omega^2/2 = 2.5521042734$.

The honest residual. The single unresolved *absolute scale* is the odd topological susceptibility $\chi_-(\mathbf{A}, \mathbf{Q}, M_p)$ (F32.27), together with **two explicitly registered conditional bridge gates**: the all-loop proof that the mixed beta function $\beta_- = 0$, and the holonomy map $\theta_Z = \omega$. We *decline* to derive χ_- parameter-free: an anti-numerology audit shows the tempting 10^{-Q^2} coincidence is a reduced- M_p + base-10 artifact, and no e-natural Z-Spin exponent lands in the required window. χ_- is registered as the *Odd Topological Susceptibility Theorem*, an executable target for a separate UV paper. $(\mathbf{A}, \mathbf{Q}, \dim Z) = (35/437, 11, 2)$
LOCKED.

Epistemic Status Legend

STATUS	DEFINITION
PROVEN	Complete proof from standard mathematics; symbol- or machine-verifiable.
VERIFIED-NUMERICAL	A finite-resolution numerical existence/robustness test, not an analytic proof.
IMPORTED-PROVEN	Proved in the external peer-reviewed literature; used with citation, not re-proved.
DERIVED	Follows from the Z-Spin structure plus standard mathematics; no new fitted parameter.
DERIVED-CONDITIONAL	Derived conditional on a stated hypothesis not itself closed here.
CLOSED-NEGATIVE	A registered route, reading, or equality resolved in the negative (a no-go).
HYPOTHESIS-strong	A specific, motivated candidate mechanism, not yet derived; falsifiable.
OBSERVATION	A numerical proximity or back-solved relation, NOT a zero-parameter prediction.
NON-CLAIM	Explicit statement of what is not asserted, to prevent overclaim.
OPEN	A recognized gap; an executable promotion route is named where one exists.
LOCKED	Core constant fixed upstream; immutable downstream.

§1. Introduction

1.1 Scope of the consolidated final version

ZS-F32 v1.3 introduced the compact-monodromy reduction but compressed the earlier theorems into a one-paragraph summary. For a standalone paper that is too terse. This version restores every theorem statement with a proof sketch and a falsification condition, and corrects four points in the B3 section flagged on review.

1.2 Corrections incorporated in v1.4

Table 1. v1.3 $\rightarrow$ v1.4 corrections.

v1.3 item	Issue	v1.4 fix
F32.1–F32.23 compressed to a summary	Not self-contained for a standalone paper	All 27 results and registered gates restored: hypotheses + conclusion + proof sketch + falsification (§2–§8)
F32.25 “IMPORTED-PROVEN branch structure”	Quadratic $E_k(0)$ is only the canonical quadratic / large-N law; general $K(F)$ allows cosine etc.	Scoped: quadratic/large-N law added as an explicit hypothesis; IMPORTED-PROVEN within that model, DERIVED-COND for the Z-Spin sector

v1.3 item	Issue	v1.4 fix
F32.26 “Friedrichs Boundary Selector”	Once circle+periodicity+holonomy ω are fixed the spectrum follows; Friedrichs does not select them	Renamed Compact-Circle Spectral Confirmation; a confirmation of F32.25, not an independent selector
Abstract “single irreducible obstruction”	Ledger itself lists $\theta_Z=\omega$ (DC) and $\beta_-=0$ (OPEN) as well	“single unresolved absolute scale χ_- + two registered conditional bridge gates”; B3 shown in three layers (§8.5)

1.3 Results Architecture (master index of F32.1–F32.27)

Exact statements are in §2–§8 (by number); this table indexes all 27 with status, principal dependency, verification item, and residual condition.

#	Theorem	Status	Dep.	Ver.	Residual / condition
1	Modular-damping no-go	PROVEN	M1	P1–2	—
2	Phase-covariant GKLS classification	PROVEN	F32.1	P3	physical branch selection
3	Principal-branch minimal action	PROVEN+DC	F32.2	P4	least-action principle
4	Seam-twisted covariance	PROVEN	F32.2	P5	—
5	Koopman–Fock completion	PROVEN+DC	F32.2	P6–7	$\dim Z = 2$
6	Finite face algebra (not MASA)	DERIVED	A30	P9	maximality NON-CLAIM
7	Canonical face expectation E_{face}	PROVEN	F32.6	P10	—
8	Wirth register lift E_{reg} (separated)	IMP+DC	A24	I1	$E_{\text{face}}=E_{\text{diag}}\circ E_{\text{reg}}$ (DC)
9	CE-local extension uniqueness	PROVEN+DC	F32.8	—	3 locality conditions
10	Entropy-production intertwining	DC	Bardet	R3	factorized lift
11	Pure-boost no-go	CLOSED-NEG	F30	P11	—
12	Loxodromic existence	PROVEN	F32.11	P12	—
13	Selective sequestering (even)	IMP+DC	OmniaSeq	—	sector separation
14	Real-de-Rham non-selection	CLOSED-NEG	HT	—	—
15	Central-shift orthogonality	PROVEN	A26	P8	—
16	Absolute-scale frontier	OPEN→F32.27	F32.15	—	$\rightarrow \chi_-$
17	Type-mismatch no-go	CLOSED-NEG	CLPW	I5	—
18	Face-embedding coarse-graining	DC	F32.6,17	—	Condition C; QRF caveat
19	Single-pulse loxodromic no-go	CLOSED-NEG	F30	P13	—
20	Two-pulse Koenigs–F30 matching	PROVEN + VERIFIED-NUMERICAL	F32.19	P14,16	—
21	Minimal angular-action pair	DC	F32.20	P14	G-7; not global opt.
22	Parity-only isolation no-go	CLOSED-NEG	F32.13	—	—
23	Flux-selection trilemma	DERIVED	F32.14,22	—	within current class
24	Compact odd 3-form protection	DC	F32.22	I6	all-loop $\beta_-=0$ OPEN
25	θ -branch selection $k^*=0$	PROVEN under C1–C2	F32.24	P17	$\theta_Z=\omega$; quad/large-N law
26	Compact-circle spectral confirmation	IMPORTED-PROVEN + DC	F32.25	P18	confirms F32.25

#	Theorem	Status	Dep.	Ver.	Residual / condition
27	Odd topological susceptibility frontier	OPEN	F32.25	audit	$\chi^-(A,Q,M_P)$

1.4 Locked inputs

$(A, Q, \dim Z) = (35/437, 11, 2)$; $z^* = i^z$, $\lambda^* = (i\pi/2)z^*$, $\kappa = 0.1148346250$, $\omega = \arg \lambda^* = 2.2592495540$ ($0 < \omega < \pi$); $\omega^2/2 = 2.5521042734$. 121 = 11² faces (ZS-A30). ZS-F30 leading support model $\beta(\theta) = \frac{1}{8} \cos 3\theta$, $\rho(\theta) = \operatorname{artanh} \beta(\theta)$, $\varphi(\theta) = \theta/2$.

§2. Local Z-Dynamics (F32.1–F32.5)

No-Go Theorem F32.1 (Modular-Damping No-Go) [PROVEN]. Let $U_t = \sigma_t \varphi$ be a real-time modular automorphism. Then U_t cannot realize the i-tetration contraction $|\lambda^*| < 1$.

Proof sketch. A modular automorphism is a *-automorphism and is isometric for the GNS norm, so its spectrum lies on the unit circle and it preserves spectral radius 1; a contracting eigenvalue $|\lambda^*| < 1$ is therefore impossible. The i-tetration map is genuinely completely-positive-dissipative, not automorphic.

Falsification. Exhibit a modular automorphism with a spectral value of modulus < 1 .

Theorem F32.2 (Phase-Covariant GKLS Classification) [PROVEN]. Every population-fixing, phase-covariant qubit generator whose time-one map sends $\sigma_+ \mapsto \lambda^* \sigma_+$ is

$$Gn(X) = (i\omega n/2)[\sigma_z, X] + (\kappa/2)(\sigma_z X \sigma_z - X), \quad \omega n = \omega + 2\pi n, \quad n \in \mathbb{Z}.$$

Proof sketch. Phase covariance forces the Hamiltonian $\propto \sigma_z$ and the dissipator to be pure dephasing; matching the time-one map fixes the modulus ($\rightarrow \kappa$) and the argument mod 2π ($\rightarrow \omega_n$). Each integer n gives the same $\exp = \lambda^*$. The v1.0 “unique generator” was thus false; the family is complete.

Falsification. A population-fixing phase-covariant generator outside this family realizing λ^* .

Theorem F32.3 (Principal-Branch Minimal Action) [PROVEN + DERIVED-CONDITIONAL].

Since $0 < \omega < \pi$, the branch $n = 0$ uniquely minimizes $\|H_n\| = |\omega + 2\pi n|/\sqrt{2}$.

$$\|H_0\| = 1.598 < \|H_{-1}\| = 2.845 < \|H_1\| = 6.040 < \dots$$

Proof sketch. $\|H_n\|$ is monotone in $|\omega + 2\pi n|$; for $\omega \in (0, \pi)$ the nearest multiple is $n = 0$. Physical selection is conditional on a least-action principle (Berkson–Porta principal-branch characterization). *Falsification.* A least-action principle selecting $n \neq 0$, or $\omega \geq \pi$.

Theorem F32.4 (Seam-Twisted Covariance) [PROVEN]. The seam $\beta = \operatorname{Ad}(\sigma_x)$ inverts the frequency:

$$\beta Pt(\omega) \beta^{-1} = Pt(-\omega).$$

Proof sketch. $\sigma_x \sigma_z \sigma_x = -\sigma_z$ flips the Hamiltonian sign while the pure-dephasing dissipator is invariant under $\operatorname{Ad}(\sigma_x)$; hence $\omega \rightarrow -\omega$. Verified $\|\operatorname{diff}\| = 0$. *Falsification.* β failing to send $\omega \rightarrow -\omega$.

Theorem F32.5 (Koopman–Fock Completion) [PROVEN + DERIVED-CONDITIONAL]. The Koopman tower $\{\lambda^* n\}$ lives in the symmetric Fock / tensor completion $\Gamma_s(H_Z)$; the local register dimension $\dim Z = \dim H_Z = 2$ is unchanged (the Fock history space itself is infinite-dimensional).

Proof sketch. The single-site time-one map sends $\sigma_+ \rightarrow \lambda^* \sigma_+$; tensor powers give λ^{*n} on $\sigma_+^{\wedge \{\otimes n\}}$; the closure is the symmetric Fock space $\Gamma_s(H_Z)$, which is infinite-dimensional as a history/excitation space but does NOT alter the local Z-register dimension $\dim Z = \dim H_Z = 2$. So the v1.0 Koopman truncation

is replaced by a completion that preserves the local register dimension. *Falsification.* The tower forcing $\dim Z \neq 2$.

§3. The Continuous-Core Operator Structure (F32.6–F32.9)

Proposition F32.6 (Finite Modular Face Algebra) [DERIVED; maximality NON-CLAIM]. The 121 equal-trace face projections q_a in the Type II₁ observer factor generate

$$A_{\text{face}} = W^*(q_1, \dots, q_{121}) \cong \mathbb{C}^{121} \quad (\text{abelian, } \sigma\text{-invariant}), \quad \text{but } A_{\text{face}} \subsetneq A_{\text{face}}' \cap M \quad (\text{NOT maximal abelian}).$$

Proof sketch. Mutually orthogonal commuting projections give an abelian algebra; the modular density $h = \sum d_a q_a$ lies in A_{face} , so it is σ_t -invariant. But in a II₁ factor each corner $q_a M q_a$ is a non-commutative algebra commuting with every q_b , so $\oplus_a q_a M q_a \subseteq A_{\text{face}}' \cap M$ strictly contains A_{face} . Hence “MASA” is withdrawn. *Falsification.* A faithful normal proof that A_{face} is maximal abelian in a II₁ factor.

Theorem F32.7 (Canonical Face Conditional Expectation) [PROVEN]. There is a unique τ -preserving conditional expectation onto A_{face} :

$$E_{\text{face}}(x) = \sum_a [\tau(q_a x q_a) / \tau(q_a)] q_a = \frac{1}{121} \cdot \sum_a \tau(q_a x) q_a.$$

Proof sketch. Verify the four CE axioms: identity on A_{face} , trace-preservation $\tau \circ E_{\text{face}} = \tau$, idempotence $E_{\text{face}}^2 = E_{\text{face}}$, modular covariance $E_{\text{face}} \circ \sigma_t = \sigma_t \circ E_{\text{face}}$ (all checked on a representative model). *Falsification.* E_{face} failing any conditional-expectation axiom.

Proposition F32.8 (Wirth Register Lift, separated from E_{face}) [IMPORTED-PROVEN + DERIVED-CONDITIONAL]. The GNS-symmetric finite-register QMS L_s lifts to the *non-commutative* register inclusion via $E_{\text{reg}} : M_{\text{obs}} \rightarrow \mathfrak{u}(M_{11})$, which is distinct from the classical face coarse-graining E_{face} .

Proof sketch. Wirth's (2022) Christensen–Evans extension provides the non-commutative QMS lift to $\mathfrak{u}(M_{11})$. E_{face} targets the abelian $\mathbb{C}^{121}$ and carries the rank/entropy budget; E_{reg} targets the non-commutative register and carries L_s . They are different maps; any relation $E_{\text{face}} = E_{\text{diag}} \circ E_{\text{reg}}$ (E_{diag} the diagonal expectation $M_{11} \rightarrow A_{\text{face}}$) is a separate DERIVED-CONDITIONAL claim. This repairs the v1.2 conflation. *Falsification.* E_{reg} shown identical to E_{face} without the diagonal map E_{diag} .

Theorem F32.9 (CE-Local Extension Uniqueness) [PROVEN + DERIVED-CONDITIONAL]. Under the three locality conditions (i) $L|N = L_s$, (ii) $\Phi = \Phi \circ E$, (iii) L has no independent Hamiltonian or dissipator on $\ker E$, the extension of L_s to M_{obs} is unique.

Proof sketch. Let $D = L_1 - L_2$ be the difference of two extensions. By (i), $D|_N = 0$; by (ii), $E \circ D = 0$; by (iii), D carries no independent generator on $\ker E$. Hence $D = 0$ and $L_1 = L_2$. Uniqueness is thus proved directly from the three locality conditions (not from a Takesaki existence theorem). Application to Z-Spin is DERIVED-CONDITIONAL on the conditions holding physically. *Falsification.* Two distinct local extensions satisfying all three conditions.

§4. Bulk–Boundary Entropy Intertwining (F32.10)

Conditional Proposition F32.10 (Entropy-Production Intertwining) [DERIVED-CONDITIONAL].

For a factorized lift $L = I_A \otimes L_{\hat{O}}$,

$$EPL = IPL + EPL\partial, \quad IP \geq 0 \Rightarrow \Sigma\partial \leq \Sigma_{bulk} \text{ (pointwise).}$$

Proof sketch. The relative-entropy production splits along the tensor factorization (Bardet et al. Prop. 5.1); the intertwining part $IP \geq 0$ by monotonicity of relative entropy, giving the boundary–bulk inequality. Verified 30/30 random states. *Falsification.* A factorized lift with $\Sigma_{\partial} > \Sigma_{bulk}$.

§5. Seam Holonomy and the Loxodromic Representation (F32.11–F32.12)

No-Go Theorem F32.11 (Pure-Boost No-Go) [CLOSED-NEGATIVE]. $\text{tr } U_Z = 0.855406 - 0.103897i$ is complex, so U_Z is not conjugate to any real pure boost.

Proof sketch. A real pure boost $\exp(\frac{1}{2} r \cdot n \cdot \sigma)$ has real trace $2 \cosh(r/2)$; $\text{tr } U_Z$ has nonzero imaginary part, so no real boost is conjugate to it. *Falsification.* A real boost with trace equal to $\text{tr } U_Z$.

Theorem F32.12 (Loxodromic Existence) [PROVEN]. U_Z is conjugate to the loxodromic element $\exp[\frac{1}{2}(-\kappa + i\omega) \cdot n \cdot \sigma]$.

Proof sketch. Its eigenvalues $e^{\pm \frac{1}{2}(-\kappa + i\omega)}$ match U_Z ; the element simultaneously contracts (rate κ) and rotates (angle ω), the defining loxodromic property in $SL(2, \mathbb{C})$. *Falsification.* An eigenvalue mismatch with $\exp[\frac{1}{2}(-\kappa + i\omega)\sigma]$.

§6. Vacuum Structure (F32.13–F32.16)

Proposition F32.13 (Selective Sequestering, even sector) [IMPORTED-PROVEN + DERIVED-CONDITIONAL]. The even vacuum-energy loops, including virtual gravitons, are removed from the curvature source by the Kaloper–Padilla *Omnia Sequestra* mechanism; the odd sector and the absolute value remain at issue.

Proof sketch. A global constraint in the sequestering action subtracts the even vacuum energy from the gravitational source at all loop orders including gravitons — an established mechanism. DERIVED-CONDITIONAL only on identifying $F_{\pm}$ with the sequestering carriers. *Falsification.* Even vacuum energy not cancelled by the mechanism, or graviton loops re-entering the source.

No-Go Theorem F32.14 (Real-de-Rham Non-Selection) [CLOSED-NEGATIVE]. A real top-form constant $f_- \in \mathbb{R}$ is a continuous free parameter; the real de Rham route does not fix it.

Proof sketch. $H^4_{-}\{dR\}(M; \mathbb{R})$ gives a continuous modulus with no integrality, so the value is unconstrained by real-de-Rham data alone. *Falsification.* A real-de-Rham argument fixing f_- .

Theorem F32.15 (Central-Shift Orthogonality) [PROVEN]. $i[H + c_0 I, X] = i[H, X]$, so operator closure is blind to the absolute offset c_0 .

Proof sketch. $c_0 I$ is central and commutes with every X , so it drops from the commutator. Verified $|\text{diff}| < 10^{-13}$. This is the A26 central-shift diagnosis: relative/modular tools cannot see the absolute vacuum value. *Falsification.* A commutator-based tool whose output depends on c_0 .

Open Gate F32.16 (Absolute-Scale Frontier) [OPEN → subsumed by F32.27]. The absolute vacuum value is orthogonal to all relative/modular tools and is registered as a frontier (the precursor to χ^-).

Proof sketch. By F32.15 the modular spectrum, relative entropy, and sector ratios fix dimensionless data but not the absolute origin; the open object is carried forward to F32.27. *Falsification.* A relative/modular tool fixing the absolute scale.

§7. Resolution of the Loxodromic Seam and the Flux Routes (F32.17–F32.23)

No-Go Theorem F32.17 (Type-Mismatch No-Go) [CLOSED-NEGATIVE]. $M_{121}(\mathbb{C})$ is a finite Type I factor; the de Sitter observer algebra is Type II₁ (Chandrasekaran–Longo–Penington–Witten). Hence the equality “corner = gravitational algebra” is impossible.

Proof sketch. Type is an isomorphism invariant: a finite Type I factor has minimal projections and a II₁ factor does not, so no normal isomorphism exists. The v1.1 “physical corner equality” is withdrawn.

Falsification. A faithful normal isomorphism $M_{121} \cong$ a II₁ factor.

Conditional Proposition F32.18 (Face-Embedding Coarse-Graining) [DERIVED-CONDITIONAL].

The correct relation is a finite abelian coarse-graining of the observer algebra:

$$\mathbb{C}^{121} \rightarrow [\Phi_{\text{face}}] M_{\text{grav}}(\text{III}) \rightarrow [E_{\text{face}}] \mathbb{C}^{121}, \quad E_{\text{face}} \circ \Phi_{\text{face}} = \text{id}.$$

Proof sketch. The face embedding and canonical expectation compose to the identity on $\mathbb{C}^{121}$. That this coarse-graining carries the gravitational entropy budget is DERIVED-CONDITIONAL on the ZS-F23 Condition C. NON-CLAIM: the gravitational algebra can depend on the observer/QRF, so an “observer-independent unique corner” may be the wrong target. *Falsification.* The coarse-graining failing to recover the (3,2,6)/11 budget under Condition C.

No-Go Theorem F32.19 (Single-Pulse Loxodromic No-Go) [CLOSED-NEGATIVE]. A single ZS-F30 pulse cannot realize $(-\kappa, \omega)$:

$$\rho(2\omega) = \text{artanh}(\% \cos 6\omega) = +0.0687857 \neq -\kappa = -0.1148346.$$

Proof sketch. With $\varphi(\theta) = \theta/2$, realizing rotation ω forces $\theta = 2\omega$, but then the rapidity $\rho(2\omega)$ has the wrong sign and magnitude. ZS-F30.7 proves generation of $\text{SL}(2, \mathbb{C})$, not single-pulse selection.

Falsification. $\rho(2\omega) = -\kappa$ for a single leading-model pulse.

Theorem F32.20 (Two-Pulse Koenigs–F30 Matching) [PROVEN + VERIFIED-NUMERICAL]. Two same-axis pulses with $\theta_1 + \theta_2 = 2\omega$ and $\rho(\theta_1) + \rho(\theta_2) = -\kappa$ compose exactly to U_Z ; two unordered families exist:

$$A: (0.8707, 3.6478), S = 14.06; \quad B: (1.5534, 2.9651), S = 11.20.$$

Proof sketch. Same-axis exponents commute and add; solving the two real equations gives the families, each verified to compose to U_Z (below numerical tolerance). A full-numerical Reuleaux-boundary test (constant width to 2×10^{-7}) again yields composing pulse-pairs, so the result is not an artifact of the leading $\cos 3\theta$ model (VERIFIED-NUMERICAL, not analytic). *Falsification.* No (θ_1, θ_2) solving both equations, or the numerical Reuleaux recomputation yielding none.

Conditional Proposition F32.21 (Minimal Angular-Action Pair within the Leading Same-Axis Two-Pulse Class) [DERIVED-CONDITIONAL]. Within the leading $\% \cos 3\theta$ same-axis two-pulse class, Family B (smaller S_{ang}) is the minimal-angular-action realization.

Proof sketch. B has the smaller angular action ($11.20 < 14.06$). DERIVED-CONDITIONAL on least-action + ZS-F30 G-7. Caveat: this proves representability, not that F30 geometry independently predicts the connection, and the minimum is within the same-axis two-pulse family, not a global optimum over axes / multi-pulse / gauge-equivalent decompositions. *Falsification.* A physical action selecting A, or a lower-action multi-pulse / different-axis decomposition.

No-Go Theorem F32.22 (Parity-Only Isolation No-Go) [CLOSED-NEGATIVE]. Since F_- is seam-odd, F_-^2 is seam-even, so $\mathbb{Z}_2$ parity alone does not force $\gamma_{+-} = 0$.

$$F_-^{-2} R, \quad F_-^{-2} F_+^2, \quad F_-^{-2} \text{ Even} \quad \text{are all parity-allowed.}$$

Proof sketch. The vacuum operator F_-^{-2} is even under the seam, so the listed mixed counterterms are seam-invariant and the cross-coupling is not forbidden by parity. All-loop isolation needs something strictly stronger (gauge superselection, a non-renormalization theorem, or a vanishing mixed beta function). *Falsification.* A symmetry argument forcing $\gamma_{+-} = 0$ from seam parity + 3-form gauge invariance alone.

Proposition F32.23 (Flux-Selection Trilemma) [DERIVED]. Real de Rham, single integral flux, and multi-flux discretuum each fail to select $|f_-|$ in the current class.

Proof sketch. Real de Rham leaves a continuous constant (F32.14); single integral flux $f_- = f_0 + nq$ leaves (q, n, f_0) and the BP step is generically too coarse; multi-flux gives a dense set but needs a measure/branch rule. This is NOT an absolute no-go (B3 is open, not impossible). *Falsification.* A current-class flux law fixing $|f_-|$ without fitting (would close B3, not falsify).

§8. The Odd-Flux Frontier: B3 = B3 protect $\wedge$ B3 select

B3 is two coupled problems — **protection** (does a small odd flux survive all-loop corrections?) and **selection** (which value?). §7 closed parity-only protection (F32.22) and the flux trilemma (F32.23). This section imports the compact-3-form / monodromy / θ -vacuum / spectral apparatus, with the scope corrected per review, and isolates the residual in three layers.

Conditional Proposition F32.24 (Compact Odd Three-Form Protection)

[DERIVED-CONDITIONAL]. Promoting F_- to the curvature of a compact $U(1)$ 3-form gauge field gives Dirac-quantized flux,

$$(1/2\pi) \int M F_- \in \mathbb{Z},$$

Proof sketch. Protection of the odd sector is then carried by the compact gauge symmetry plus a discrete axion shift (Kaloper–Sorbo / four-form monodromy), NOT by $\mathbb{Z}_2$ parity (which fails, F32.22). The compact-flux mathematics and the monodromy EFT radiative control are IMPORTED-PROVEN; the identification of Z-Spin $F_\pm$ with these carriers is DERIVED-CONDITIONAL; complete all-loop graviton-mixed isolation ($\beta_{+-} = 0$) remains OPEN. “Radiative control” is not the same as “all-loop isolation.” *Falsification.* A no-go forbidding a compact 3-form structure for F_- , or showing monodromy gives no radiative control.

Conditional Proposition F32.25 (θ -Branch Selection) [PROVEN under C1–C2]. Assume **(C1)** the canonical quadratic / large-N branch law and **(C2)** the holonomy identification $\theta_Z := \arg \lambda^* = \omega$. Then $0 < \omega < \pi$ selects $k^* = 0$ uniquely:

$$Ek(\theta) = (\chi^-/2)(\theta + 2\pi k)^2, \quad k^* = 0, \quad \rho A_Z = (\chi^-/2) \omega^2 = 2.5521042734 \cdot \chi^-.$$

Proof sketch. Under the quadratic law one minimizes $(\theta + 2\pi k)^2$ over $k \in \mathbb{Z}$; for $\theta = \omega \in (0, \pi)$ the minimizer is $k = 0$ since $|\omega| < |\omega \pm 2\pi|$. SCOPE (correction): the quadratic $E_k(\theta)$ is the canonical / large-N law (Witten large-N θ -dependence); a general kinetic $K(F)$ can give a cosine or other potential, so the law is IMPORTED-PROVEN only within that model and DERIVED-CONDITIONAL for the Z-Spin odd sector. C2 is DERIVED-CONDITIONAL (the BV–BFV / anomaly-inflow map $U(1)_Z \rightarrow 3\text{-form } \theta\text{-angle}$

is unproved). Given C1 and C2, $k^* = 0$ and the coefficient $\omega^2/2$ are PROVEN. This is a branch selection, not a new selection of the absolute value. *Falsification.* A non-quadratic $K(F)$ effective potential for the Z-Spin sector, or $\theta_Z \neq \omega$, or $\omega \geq \pi$.

Proposition F32.26 (Compact-Circle Spectral Confirmation) [IMPORTED-PROVEN + DERIVED-CONDITIONAL]. On the compact flux circle $\phi \sim \phi + 1$, the semibounded form

$$q\omega[\psi] = (\chi-/2) \|(-i d/d\phi + \omega)\psi\|^2, \quad \text{spectrum } pk = (\chi-/2)(2\pi k + \omega)^2, \quad \text{ground } k = 0.$$

Proof sketch. On the circle $-i d/d\phi$ has eigenvalues $2\pi k$; the form is semibounded and its Friedrichs realization gives the stated spectrum (Reed–Simon X.3), with ground state $k = 0$ for $0 < \omega < \pi$. NAME (correction): this CONFIRMS F32.25 spectrally — it does not independently select why the flux space is a circle, why the bundle is periodic, or why the holonomy is ω ; those are inputs from F32.24/C2. It is therefore a confirmation, not an independent selector. *Falsification.* The spectrum differing from $(2\pi k + \omega)^2$, or the result being read as an independent boundary selector.

Open Problem F32.27 (Odd Topological Susceptibility Frontier) [OPEN]. The single unresolved absolute scale is $\chi_-(A, Q, M_P)$, with the conditional law

$$\rho_{\Lambda,Z} = \frac{1}{2} \chi_- \omega^2 = 2.5521042734 \cdot \chi_-.$$

Proof sketch. We decline to derive χ_- parameter-free (anti-numerology): the observed value needs $\ln(M_P^4/\chi_-) \approx 277.6$ (reduced M_P) vs 284.0 (full M_P) — convention-dependent. The clean $10^{-(Q^2)} = 10^{-121}$ matches only in reduced- M_P + base-10 (full M_P off by $\sim 10^{2.8}$; $\ln 10$ is not a Z-Spin quantity). No e-natural exponent ($8\pi^2/A=986$, $8\pi^2=79$, $Q^2=121$, $Q^2\omega=273$, $\pi/A=39$) lands in $[277.6, 284.0]$. Back-solving would be hidden fitting, which we refuse. Named UV candidates (electroweak-instanton susceptibility; q-theory eighth power $\chi_- \sim \Lambda_O^8/M_P^4$, dimensionally = A28 double-see-saw; A-transmutation) are HYPOTHESIS-strong, none deriving χ_- blind from the locked constants. *Falsification.* Any claim that χ_- is derived parameter-free using observed ρ_Λ , a chosen scale, a base-10 coincidence, or a back-solved exponent is void.

8.5 The residual in three layers, and the registered frontier

Table 2. The honest final state of B3 (three layers).

Layer	Object	Status
B3 bridge-1	holonomy map $\theta_Z = \omega$ ($U(1)_Z \rightarrow 3$ -form θ -angle)	DERIVED-CONDITIONAL
B3 bridge-2	all-loop odd-sector isolation $\beta_- = 0$	OPEN
B3 scale	absolute susceptibility $\chi_-(A, Q, M_P)$	OPEN

So the cosmological residual is **one unresolved absolute scale (χ_-) plus two registered conditional bridge gates**, not a single object. B3 compresses to one executable programme — the *Odd Topological Susceptibility Theorem* — with four gates (no observed input permitted): **B3-1** derive $F_- \in H^*(M; \mathbb{Z})$ and the charge lattice from the axioms; **B3-2** a BV–BFV / BRST counterterm-cohomology proof that $\beta_- = 0$ to all orders; **B3-3** supplied by F32.25–F32.26 under C1–C2 ($k^* = 0$); **B3-4** output ρ_Λ before comparison with data. Only then does B3 become DERIVED.

§9. Conclusion

ZS-F32 does not derive the observed cosmological constant. It proves and constructs the operator-algebraic, Markovian, modular, and loxodromic structures required to reduce the absolute-scale problem to a UV odd-sector susceptibility. Under a compact quadratic branch model and the conditional holonomy identification $\theta_Z = \omega$, the integer branch is uniquely $k^* = 0$, yielding $\rho_{\Lambda Z} = \frac{1}{2} \chi^- \omega^2$. The all-loop isolation of the odd sector ($\beta_- = 0$), the holonomy map $\theta_Z = \omega$, and the parameter-free derivation of χ^- remain open.

This closes the continuous-core operator and seam structure as tightly as the present toolset allows, converts “why is Λ small?” into the precise falsifiable task “compute χ^- parameter-free, then blind-predict ρ_{Λ} ,” and — via the anti-numerology audit — declines to fabricate that magnitude. The remaining work (UV gauge group, odd matter content, charge lattice, instanton action, mixed-counterterm cohomology) is the scope of a separate UV field-theory paper, ZS-F33, for which F32 provides the exact starting point. (A, Q, dim Z) = (35/437, 11, 2) **LOCKED**.

Acknowledgements & Code Availability

This consolidated version responds to a review of v1.3 requesting restored self-containment and four B3-scope corrections (quadratic/large-N branch law, the spectral-confirmation relabel, the three-layer residual, and the abstract framing). The branch selection ($k^* = 0$ from $\theta_Z = \omega$, $0 < \omega < \pi$), the compact-circle spectrum, and the χ^- anti-numerology audit were checked numerically (numpy / scipy / mpmath, seed 437). **Provenance and the refusal to fit.** Per ZS-A28 v2.3, internal consistency is not external validation. χ^- is *not* derived here; $\rho_{\Lambda} = \frac{1}{2} \chi^- \omega^2$ is a falsification target, and the observed value is not used as an input. PROVEN / IMPORTED-PROVEN results rest on standard theorems (Chandrasekaran–Longo–Penington–Witten; Takesaki; Wirth 2022; Bardet et al.; Kaloper–Sorbo / Kaloper–Padilla; Witten large-N θ ; Bousso–Polchinski; Reed–Simon). B3 remains OPEN — not closed, not proven impossible (ZS-A26). Reproducibility scripts (zs_f32_verify_v1_5.py) are available at the Z-Spin Collaboration repository.

Appendix A. Verification Ledger (by category)

Table A1. PROVEN / VERIFIED checks (18).

#	Check	Result
P1–P2	κ, ω ($0 < \omega < \pi$)	0.1148346250 ; 2.2592495540
P3–P5	branch independence; norm-min $n=0$; seam inversion	verified
P6–P7	M_2 spectrum (no λ^{*2}); Fock tower λ^{*n}	verified
P8	central-shift orthogonality	$\ \text{diff}\ < 1e-13$
P9–P10	finite face algebra abelian + σ_t -inv (NOT MASA, witness); E_{face} canonical	NON-MASA witness + representative
P11–P12	pure-boost no-go (complex tr); loxodromic match	verified
P13–P14	single-pulse no-go; two-pulse composition = U_Z	verified
P15	Reuleaux-boundary robustness (numerical)	VERIFIED-NUMERICAL
P16	two-pulse constraints $\Sigma \rho = -\kappa$, $(\theta_1 + \theta_2)/2 = \omega$	$1e-9$
P17	θ -branch $k^*=0$ under $(\theta_Z = \omega \wedge \text{quadratic law})$; $\omega^2/2$	2.5521042734 ($0 < \omega < \pi$)
P18	compact-circle spectrum $(2\pi k + \omega)^2$ ground $k=0$ (confirms P17)	$\omega < 2\pi - \omega$

Table A2. IMPORTED-PROVEN theorem-conditions (7).

#	External theorem	Role
I1–I4	Wirth 2022; Bardet et al.; Poincaré/HT/Bousso–Polchinski; Berkson–Porta	F32.8, F32.10, F32.14/23, F32.3
I5	Chandrasekaran–Longo–Penington–Witten — de Sitter observer algebra Type II ₁	F32.17
I6	Compact U(1) 3-form Dirac quantization; Kaloper–Sorbo monodromy; Witten large-N θ (quadratic law)	F32.24, F32.25
I7	Reed–Simon — Friedrichs extension of a semibounded form; min–max	F32.26

Table A3. Anti-numerology audit of χ^- (why it is NOT closed).

Attempt	Finding	Verdict
$10^{-(Q^2)}=10^{-121}$	reduced-M_P+base-10 only; full M_P off by $\sim 10^{2.8}$; ln10 not a Z-Spin quantity	REJECTED (convention artifact)
transmutation b-	$b- \approx 13.9\text{--}14.2$ (convention-dependent); needs odd matter content	OPEN (not derivable)
natural e-exponents	$8\pi^2/A=986$, $8\pi^2=79$, $Q^2=121$, $Q^2\omega=273$, $\pi/A=39$ — none in [277.6,284.0]	no parameter-free hit
eighth power $\Lambda_{\text{O}}^8/M_{\text{P}}^4$	dimensionally OK (= A28 double-see-saw); Λ_{O} ($\sim$ TeV) not fixed by A,Q	OPEN (scale free)

Appendix B. Falsification Gates

Gate	Layer	Condition
F-F32.2/6	Mathematical	A generator outside the F32.2 family realizing λ^* ; or A_face proven maximal abelian in a II ₁ factor.
F-F32.8	Mathematical	E_reg shown identical to E_face without the diagonal map.
F-F32.17	Mathematical	A normal isomorphism $M_{121} \cong$ a II ₁ factor (impossible by type).
F-F32.20	Mathematical/Numerical	No (θ_1,θ_2) solving both equations, or the numerical Reuleaux test yielding none.
F-F32.25	Physical	A non-quadratic K(F) potential for the Z-Spin sector, $\theta_Z \neq \omega$, or $\omega \geq \pi$.
F-F32.27	Meta / anti-numerology	Any parameter-free χ^- claim using observed $\rho\Lambda$, a chosen scale, a base-10 coincidence, or a back-solved exponent is void.
F-B3	Theoretical	A Z-Spin-internal derivation of $\chi^-(A,Q,M_P)$ passing B3-1..B3-4 would CLOSE B3 (not falsify).

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Version History

v1.0–v1.2 (March–June 2026): v1.0 modular-damping no-go and Wirth lift; v1.1 corrected the false GKLS uniqueness, the Koopman truncation, and the topological over-claim, adding seam-twist, Gate-E, and loxodromic existence; v1.2 repaired the face-corner (Type I vs II_1), constructed the two-pulse loxodromic seam, closed parity-only isolation, and recorded the flux trilemma.

v1.3 (June 2026): Repaired the “Face-MASA” to a finite non-maximal abelian face algebra and separated E_{reg} from E_{face} ; relabeled the Reuleaux test VERIFIED-NUMERICAL; advanced B3 with compact-3-form quantization, monodromy protection, the θ -vacuum branch with $\theta_{\text{Z}}=\omega$ selecting $k^*=0$, and the boundary selector; isolated χ^- with an anti-numerology audit declining to fit it.

v1.4 (June 2026, consolidated final): Restored full self-containment — all 27 theorems (F32.1–F32.27) restated with hypotheses, conclusion, proof sketch, and falsification condition, plus a single Results-Architecture index. Corrected the B3 section: F32.25 scoped to the canonical quadratic / large-N branch law (IMPORTED-PROVEN within that model, DERIVED-CONDITIONAL for the Z-Spin sector) and stated as PROVEN only under $(\theta_{\text{Z}}=\omega \wedge \text{quadratic law})$; F32.26 renamed Compact-Circle Spectral Confirmation (a confirmation of F32.25, not an independent selector); the residual presented in three layers — $\theta_{\text{Z}}=\omega$ (DERIVED-CONDITIONAL), $\beta_+-=0$ all orders (OPEN), $\chi^-(A, Q, M_P)$ (OPEN); abstract reworded to “one unresolved absolute scale + two conditional bridge gates.” Strengthened references (Witten large-N θ ; Marchesano–Shiu–Uranga general kinetic functions; Kaloper–Padilla one-loop). No numerical claim changed. $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

v1.5 (June 2026, corrected final): Editorial and rigor corrections, no content removed and no numerical claim changed: fixed the F32.3 branch-index ordering ($\|H_0\| < \|H_{-1}\| < \|H_1\|$ since $\omega_{-1}=\omega-2\pi$); relabeled the F32.2 family “integer-indexed (countable logarithmic-branch)” rather than “one-parameter”; restored the three F32.9 locality conditions as equations and proved uniqueness directly via $D=L_1-L_2$ (removing the imprecise Takesaki attribution); fixed the Results-Architecture verification link for F32.17 (I5, a type-classification theorem, not the P11 pure-boost check); strengthened the P9/P10 code to test per-projection modular invariance, modular covariance, and the bimodule property; sharpened the F32.5 Fock statement (the history space is infinite-dimensional; only the local register $\dim Z = \dim H_{\text{Z}} = 2$ is

preserved); narrowed F32.21 to “within the leading same-axis two-pulse class”; renamed OPEN items as Open Gate/Open Problem and applied a status-consistent naming scheme (Theorem / Proposition / Conditional Proposition / No-Go Theorem / Open) across F32.1–F32.27; corrected reference [11] (PRL 116, 051302 (2016) and the graviton-loop PRL 118, 061303 (2017)); and applied minor symbol fixes (10^{-Q^2} , PROVEN + VERIFIED-NUMERICAL, PROVEN under C1–C2). F32 is hereby concluded; $\theta_{Z=\omega}$, $\beta_{\cdot-}=0$, and χ_{-} are deferred to ZS-F33. (**A**, **Q**, $\dim Z$) = (35/437, 11, 2) LOCKED.